A Matter of Perspective: Unpacking the Effect of Different Variables on a Virtual Reality Game

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Summary: This work develops a virtual reality game whose parameters can be changed in a controlled way between sessions, in order to understand how they affect the way people experience and react to the game. The same VR game can be engaging for one player and uncomfortable for another, but it is still unclear which design choices cause these differences. Knowing the effect of each parameter supports better-informed decisions when designing VR experiences.
The parameters cover the environment where the game takes place, the way it is presented to the player, the characters the player interacts with and the task itself, and can be adjusted individually or combined. Their effect is measured through in-game performance and questionnaires. The game takes place in an industrial environment, where the player is a maintenance technician who must diagnose and fix a broken machine.

Objectives / Work Done / Results

Objectives

- Map the parameters that can influence the player experience in VR
- Design and develop a VR game whose parameters can be manipulated across sessions
- Evaluate the VR game with users under different configurations and understand which parameters affect the player experience



Fig 1:Reference-I Expect You To Die

Work Done

- Literature review and analysis of the state of the art
- Analysis of different VR and non-VR games
- Early draft of the Game Design Document

Results

- Initial concept of the game, mechanics and parameters

Next Steps / Challenges / Bottlenecks

Next Steps

- Continue exploring the state of the art
- Further develop the initial concept of the game
- Decide the architecture to be used
- Start the development of the first prototype
- Try to establish a partnership with a company to link the game to a case study

Challenges

- Decide a reasonable number of parameters
- Define a feasible number of experimental conditions
- Multiplayer integration
- Anxiety and cybersickness management

Bottlenecks

- Recruit users to test the parameters under analysis